

4. Nuclear reactor 3 at the Hunterston B power station was shut down after cracks in graphite moderator bricks were found to be growing faster than expected. The Office for Nuclear Regulation (ONR) raised concerns over the number of cracks in the joints between the graphite moderator bricks in the core. An unusual event, such as an earthquake, might move the graphite bricks so the control rod channels could become blocked.

- (a) (i) Explain the purpose of the graphite moderator in a nuclear reactor. [2]

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- (ii) Explain why blocked control rod channels are very dangerous. [3]

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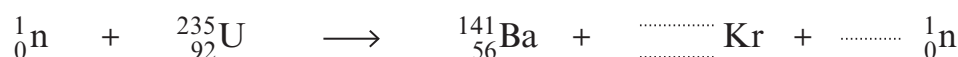
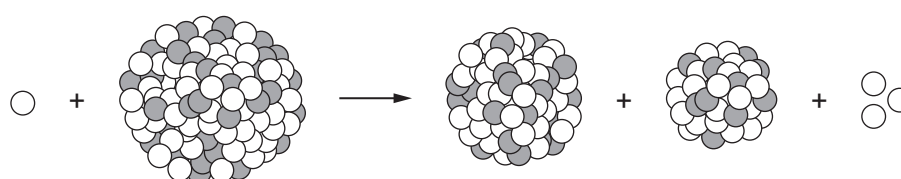
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- (b) The diagram shows a reaction in a nuclear reactor and part of the reaction equation.

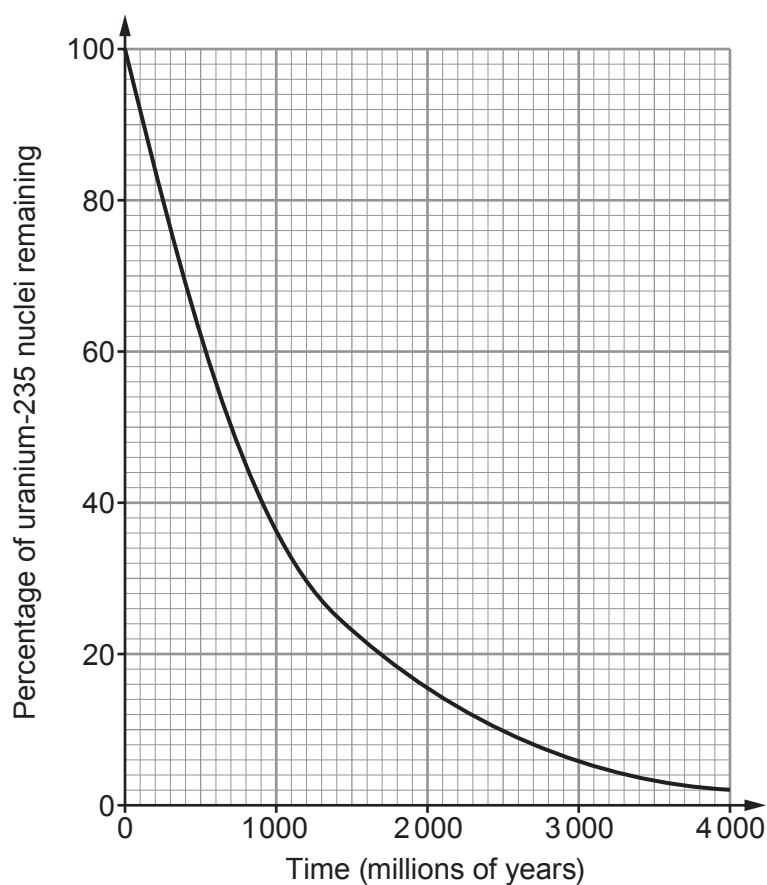


Complete the equation.

[3]



- (c) It is likely that Hunterston B power station will be decommissioned. When it is, the uranium-235 in the fuel rods will continue to decay at the rate shown in the graph below.



- (i) Add lines to the graph to calculate the half-life of uranium-235.

[2]

half-life = million years

- (ii) Uranium-235 becomes safe once its activity drops to $1/32$ of its original value. Calculate the time it takes to drop to this value.

[3]

time = years

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